

N-PQMS Phase 1 — Business Requirements Document (BRD)

TSB Publication Management · Issue Management · QIR Module · Admin Module

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Version	Date	Author	Change Summary
1.0	May 2026	Joon Sung Yoo	Initial draft
1.1	June 2026	Joon Sung Yoo	Added §7.2 Issue Source Channels (Warranty, Weibull, Comeback, Techline, FPQR); updated ISM-FR-002, ISM-FR-010, ISM-FR-011; renumbered prior §7.2-§7.9 → §7.3-§7.10; updated §5.2 ISM screen inventory and §20.3 M3 acceptance criteria

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1. Executive Summary

Phase 1 of the N-PQMS Re-Platform delivers the **TSB Publication Management (PUM)**, **Issue Management (ISM)**, and **QIR Module (QIM)** — the three modules that account for **98.7% of all current KPQMS usage** (27,938 usage hits, Jan-Nov 2025). These modules go live on **December 18, 2026**, replacing the legacy TOBESOFT XPLATFORM / Oracle WebLogic stack.

Item	Detail
Phase 1 Go-Live	December 18, 2026
Phase 1 Budget	\$824,245 (M1-M6 of the \$1,240,000 total)
Screens Delivered	54 screens (Tier 1: 20 + Tier 2: 10 + Tier 3: 24)
Usage Coverage	99.2% of all KPQMS screen usage
Core Integrations	INT-01 through INT-06
Workflow Engine	Camunda BPM (TSB approval + QIR assignment chains)
Authentication	Azure AD SSO / JWT / RBAC / MFA
Analytics	INT-06 outbound push to CDO (PQMS as data producer)
NFR Anchor Screen	PUM0020 Publication Entry — ≤2s response at 10 concurrent users

2. Business Objectives

2.1 Primary Objectives — Phase 1

#	Objective	Success Measure
B1	Replace XPLATFORM/XFDL-based TSB workflow with modern Vue.js SPA on AWS EKS	All 6 PUM screens functional at go-live
B2	Modernize QIR/Issue lifecycle management with AI-assisted severity scoring (prototype)	All 14 ISM + 10 QIM screens functional; severity score visible on QIR
B3	Retire all direct JDBC connections for Phase 1 systems; replace with REST adapter layer	INT-01–06 all functional; zero active JDBC connections in production
B4	Implement enterprise-grade auth — Azure AD SSO, RBAC, MFA — replacing LDAP/JNDI	SSO login functional; all 8 RBAC roles enforced; MFA active
B5	Activate Camunda BPM engine for TSB approval and QIR assignment workflows	BPM workflows functional; legacy <code>T_APRO_*</code> tables retired for Phase 1 scope
B6	Enable CDO analytics pipeline (INT-06 outbound) from go-live day	INT-06 confirmed by CDO sign-off at M7 (Phase 2 Dev)
B7	Surface EWS Early Warning flags (<code>EWS_FG</code>) in QIR/Issue screens	<code>POST /api/ews/signals</code> endpoint live; <code>EWS_FG</code> visible in ISM/QIM screens
B8	Establish baseline reporting: 12 standard reports + 6 ad-hoc report structure	Report structure deployed on non-prod at M3; confirmed at UAT
B9	Deliver full system administration capability — user management, master data, workflow config, system settings — in Phase 1, removing any gap dependency on a later phase for basic operations	All 33 ADM screens functional at Phase 1 go-live

2.2 Business Context

The legacy KPQMS TSB publication process is fragmented across XPLATFORM screens, manual email chains, and the decommissioned Spotfire/EWS system. KUS PQ engineers manage ~16,970 TSB-related screen interactions monthly under a framework that cannot integrate with Azure AD, Camunda BPM, or SAS Visual Analytics.

Phase 1 resolves this by:

- Consolidating the TSB lifecycle (draft → review → approval → posting → time tracking) into a single Camunda-orchestrated workflow
- Bringing QIR creation and management onto the same cloud-native platform with shared VIN/dealer data from INT-01 (AS400) and INT-03 (Siebel)
- Enabling the KUS PQ team's SAS VA dashboards via INT-06 outbound data from go-live
- Delivering the full Admin Module (ADM) at go-live so user management, master data, workflow templates, and system configuration are self-service from day one — no dependency on a later phase for administrative operations

3. Stakeholders & Counterpart Teams

3.1 Core Team

Name	Organization	Role
Kim Yoonsoo	HAEA	Project Sponsor / Primary Approver (M1-M6)
Yoo Joon Sung	HAEA	Project Manager
Ahn Hyo Jung (Jenny)	HAEA	Sr. Manager After Sales · Escalation L2
Rita Verma	Tenarai	Provider Program Manager
[TBD at M1]	Tenarai	Technical Lead
[TBD at M1]	Tenarai	Integration Lead

3.2 Phase 1 Counterpart Teams (HAEA Obligations)

CP #	Team	System	Availability	Access Deadline
CP-1	GQIS HQ Korea Integration	INT-02 GQIS (Groups A + B)	80-120h total; 12h timezone window; 2-week parallel write	Jul 3, 2026 (API in DEV)
CP-2	Vital Insights / CSI	INT-12 IQS/VDS (HTTP POST decision)	Decision + 20-40h if HTTP agreed	Jul 3, 2026 (written decision)
CP-3	HISNA / AS400 Admin	INT-01 Group C (13 endpoints)	16-24h total	Jun 1, 2026 (ACC-07)
CP-4	Active Directory / IT Admin	Group G Auth (HKE.LOCAL)	8-16h total	Jun 1, 2026 (ACC-08)
CP-7	WPC SQL Server DBA	Group F WPC FixedOps (2 endpoints)	8-12h total	Jun 1, 2026 (ACC-09)
CP-8	SMTP / Mail Relay Admin	Group H Mail Notification	4-8h total	Jun 1, 2026 (ACC-10)
CP-9	Network / Infra Admin	All groups (Adapter Service)	24-40h total	Jun 10, 2026 (N-GWMS)
PQMS Business SMEs	KUS PQ Team	All functional modules	8h/week during BRD workshops	May-Jun 2026

4. Phase 1 Scope Boundary

4.1 What Phase 1 Delivers

Module	Code	Tier	Screens	Usage
Publication Management (TSB)	PUM	1 — Critical	6	60.7%
Issue Management / QIR	ISM	1 — Critical	14	24.4%
QIR Module	QIM	2 — Important	10	13.6%
Admin Module	ADM	3 — Low	24	0.5%
Phase 1 Total			54	99.2%

Core integrations: INT-01 · INT-02 · INT-03 · INT-04 · INT-05 · INT-06

Workflow engine: Camunda BPM (TSB approval + QIR assignment + escalation)

Auth: Azure AD SSO · JWT · RBAC (8 roles) · MFA

EWS: `POST /api/ews/signals` writeback endpoint · `EWS_FG` flag in ISM/QIM screens

Reporting: 12 standard reports + 6 ad-hoc structure · prototype KPI dashboard

Data migration: Phase 1 dry-run only (INT-11 full migration is Phase 2)

4.2 What Phase 2 Delivers (for reference)

INT-07–12 (PQPS, TechLink, Camunda external API, AWS SES/SNS, legacy migration, IQS/VDS), full AI/ML 24 models, NLP/RAG intelligent search, historical data migration (INT-11), enhanced analytics, ADM ML Model Management screens (ADM0210–ADM0250) and IQS/VDS screens (ADM0260–ADM0290) — go-live March 12, 2027.

4.3 Decommissioned Systems (Not Replaced in Phase 1 or Phase 2)

System	Reason
GBI Oracle Data Warehouse (<code>jdbc/GBIBiz</code>)	Decommissioned at enterprise level
Spotfire / EWS Oracle (<code>jdbc/SPOTBiz</code>)	Decommissioned — EWS replaced by SAS/QC on CDO
S+ Statistical Analytics	Decommissioned — replaced by SAS/STAT
SPT Spotfire Analytics screens (17)	Do not migrate — users redirected to SAS VA
PUP / ISP / QIP legacy preview screens (17)	Do not migrate — absorbed into unified TSB/QIR workflows

5. Screen Inventory

5.1 Tier 1 — PUM Publication Management (6 screens)

Screen ID	Screen Name	Description	Usage Rank
PUM0020	Publication Entry	Create, edit, submit TSB publication with metadata and attachments	#1 — 30% of all system hits
PUM0010	Publication List	Search, filter, and list TSB publications by status, model, date	Top 10
PUM0030	Publication Detail	Full-view read of a TSB publication with version history and comments	Top 10
PUM0040	Publication Approval	Multi-stage approval action screen (Camunda task form)	Top 10
PUM0050	TSB Posting	Manage posting status and automate publishing to downstream channels	Top 10
PUM0060	Publication Time Tracking	Track time spent on TSB authoring, review, and approval stages	—

5.2 Tier 1 — ISM Issue Management (14 screens)

Screen ID	Screen Name	Description
ISM0010	Issue List	List and filter active issues by severity, model, system, status, and source channel (Warranty / Weibull / Comeback / Techline / FPQR / EWS / GQIS)
ISM0020	Issue Entry	Identify and log new quality issues; captures mandatory Issue Source field (Warranty / Weibull / Comeback / Techline / FPQR / EWS / GQIS / Manual) with channel-specific structured fields per §7.2
ISM0030	Issue Scoring	Score issue severity (manual prototype; AI-assist in Phase 2)
ISM0040	Issue Detail	Full-view issue record with VIN data, dealer info, and communication log
ISM0050	QIR Creation	Initiate a Quality Issue Report from a scored issue
ISM0060	QIR Assignment	Assign QIR to QE/TE/DE/CE role; trigger Camunda task
ISM0070	Issue Disposition	Record final disposition (TSB / SA / SC / No Action)
ISM0080	Issue Tracking	Track issue state transitions and milestone dates
ISM0090	Parts Request	Initiate and track part requests tied to an issue
ISM0100	Communication Log	Capture and view communication threads per issue
ISM0110	Escalation Management	Escalate issue to QIR or Top Issue status
ISM0120	Cross-Org Visibility	Share issue data across organizational units
ISM0130	EWS Flag View	Surface EWS_FG alert flags from SAS/QC via writeback endpoint
ISM0140	Issue Admin / Batch	Manage batch jobs, system admin settings for ISM module

5.3 Tier 2 — QIM QIR Module (10 screens)

Screen ID	Screen Name	Description
QIM0010	QIR List	List and search all QIR records by status, model, system, assignee
QIM0020	QIR Detail	Full QIR record view with linked ISM issue, FPQR, warranty claims
QIM0030	QIR Approval Workflow	Multi-level QIR approval action screen (Camunda task form)
QIM0040	FPQR Submission	Submit Field Product Quality Reports linked to a QIR
QIM0050	FPQR Management	Manage and track FPQR lifecycle
QIM0060	VIN Search	Search vehicle records by VIN across INT-01 (AS400) and INT-03 (Siebel)
QIM0070	Warranty Data Analysis	Analyze warranty claims data linked to QIR records
QIM0080	QIR Effectiveness Analysis	Evaluate effectiveness of QIR corrective actions
QIM0090	Campaign Reporting	Analyze campaign-related quality data
QIM0100	Issue Analytics Dashboard	Visual dashboard for issue trends, KPIs, model line / system-level breakdowns

5.4 Tier 3 — ADM Admin Module (33 screens)

User & Access Management (5 screens)

Screen ID	Screen Name	Description
ADM0010	User List & Search	List, search, and filter all N-PQMS users by role, status, org unit
ADM0020	User Entry & Edit	Create and edit user profiles: name, email, org, role assignment, access expiry
ADM0030	Role Assignment	Assign and revoke RBAC roles per user; bulk assignment via XLSX import
ADM0040	Access Audit Log	Read-only log of all user access changes — grantee, grantor, role, timestamp
ADM0050	External User Access	Manage external (non-KUS) user accounts with expiry date and role restrictions

System Configuration (5 screens)

Screen ID	Screen Name	Description
ADM0060	System Parameters	Global config: pagination defaults, file size limits, token TTL, date formats
ADM0070	Notification Template Config	Create and edit email/push notification templates per trigger event
ADM0080	Integration Status Monitor	Real-time status of all 6 Phase 1 INT workstreams; error counts, last-sync timestamps
ADM0090	Environment Config	Environment-level settings: Redis TTL, batch schedules, log retention
ADM0100	Tenant Configuration	Multi-tenant company code settings

Workflow Template Management (4 screens)

Screen ID	Screen Name	Description
ADM0110	TSB Workflow Template Editor	Create and edit Camunda BPMN templates for TSB approval (standard, expedited, emergency)
ADM0120	QIR Workflow Template Editor	Create and edit QIR assignment and approval workflow templates
ADM0130	SLA Configuration	Set SLA deadlines per workflow stage and escalation rules
ADM0140	Escalation Rules Config	Define escalation chains: who is notified when, thresholds, override rules

Master Data Management (6 screens)

Screen ID	Screen Name	Description
ADM0150	Model / Model Year Master	Manage vehicle model and model year records (INT-01 sync + manual override)
ADM0160	Plant Master	Manage production plant codes and associated regions
ADM0170	Parts Master Sync	Trigger and monitor INT-04 SAP BW parts master sync; view last-sync status
ADM0180	Op Code Master	Manage labor op codes sourced from INT-01; manual additions
ADM0190	Dealer Master Sync	Trigger and monitor INT-03 Siebel dealer master sync
ADM0200	Symptom / Cause / Correction Codes	Manage standardized S/C/C code library used across ISM/QIM/FPQR

Batch & System Health Management (4 screens)

Screen ID	Screen Name	Description
ADM0300	Batch Job Monitor	Monitor all scheduled batch jobs: status, last-run, next-run, error log
ADM0310	Migration Status Dashboard	Phase 1: dry-run tracking; Phase 2: full INT-11 migration progress
ADM0320	Data Quality Report	View data quality results from INT-06 EDW push and migration dry-run
ADM0330	System Health Dashboard	AWS CloudWatch summary: EKS pod health, RDS CPU/connections, Redis hit rate

ADM Total: 24 screens (5 + 5 + 4 + 6 + 4)

ADM0210–ADM0290 (ML Model Management and IQS/VDS Management, 9 screens) are delivered in Phase 2 when AI/ML models and INT-12 are available. They are not in Phase 1 scope.

6. Functional Requirements — TSB Publication Management (PUM)

6.1 TSB Draft Creation & Collaboration

ID	Requirement	Priority
PUM-FR-001	Users shall create a new TSB draft by entering: title, TSB type, affected model(s), affected model year(s), related symptoms, parts/op codes, and attaching supporting files	P1
PUM-FR-002	External teams (outside KUS PQ) shall be able to submit draft TSBs through a controlled submission interface with role-based access restrictions	P1
PUM-FR-003	Multiple reviewers shall be able to add comments, annotate, accept, or reject a draft TSB within the same screen session	P1
PUM-FR-004	Draft TSBs shall support version history — each save/submit action creates a versioned snapshot with author and timestamp	P1
PUM-FR-005	Draft TSBs shall auto-save at configurable intervals (default: every 5 minutes)	P2

6.2 Publication Entry & Metadata (PUM0020 — NFR Anchor)

ID	Requirement	Priority
PUM-FR-010	PUM0020 shall allow entry of all publication metadata: TSB number, revision, model, model year, system, symptom, cause, correction, parts, op codes, labor time	P1
PUM-FR-011	PUM0020 shall support file attachments: PDF, DOCX, XLSX, images — max file size and allowed types configurable by admin	P1
PUM-FR-012	PUM0020 shall integrate with INT-01 (AS400/HISNA) to auto-populate VIN master, model, plant, op code data in dropdowns	P1
PUM-FR-013	PUM0020 shall integrate with INT-04 (SAP BW/4HANA) to auto-populate parts master data	P1
PUM-FR-014	PUM0020 page load shall meet the SPEC-PERF-001 NFR target: ≤2 seconds at 10 concurrent users (DEV), ≤1.5 seconds at 50 concurrent users (Staging)	P1
PUM-FR-015	Publication number shall be auto-generated per configurable numbering scheme; manual override permitted for Admin role	P1

6.3 TSB Approval Workflow

ID	Requirement	Priority
PUM-FR-020	TSB shall follow a configurable multi-stage approval chain managed by Camunda BPM: Draft → Technical Review → Manager Approval → Director Approval → Published	P1
PUM-FR-021	Each approval stage shall generate an automated notification (email via Group H) to the next approver with a direct link to PUM0040	P1
PUM-FR-022	Approvers shall be able to approve, reject with comments, or return to author at any stage	P1
PUM-FR-023	Rejected TSBs shall revert to Draft status with the rejection reason visible in the version history	P1
PUM-FR-024	Approval deadline per stage shall be configurable; overdue tasks shall trigger escalation notification to the stage supervisor	P2
PUM-FR-025	Parallel approval paths shall be supported for stages requiring sign-off from multiple roles simultaneously	P2
PUM-FR-026	TSB approval workflow templates shall be configurable by Admin role without code changes	P2

6.4 Publication Listing & Search

ID	Requirement	Priority
PUM-FR-030	PUM0010 shall support search and filter by: TSB number, title, status, model, model year, system, date range, author	P1
PUM-FR-031	Search results shall be paginated (default 20 per page; configurable) with sort by date, model, status	P1
PUM-FR-032	Publication status shall be displayed as a color-coded badge: Draft, In Review, Approved, Published, Archived	P1
PUM-FR-033	Users shall be able to export search results to XLSX	P2

6.5 TSB Posting & Publishing Management

ID	Requirement	Priority
PUM-FR-040	PUM0050 shall track publishing status: pending, posted, confirmed, archived	P1
PUM-FR-041	Publishing status updates shall integrate with INT-08 (TechLink) for DTC/TSB cross-reference — <i>Note: INT-08 is Phase 2; Phase 1 shall use manual status entry with INT-08 stub</i>	P2
PUM-FR-042	Posting confirmation shall be logged with timestamp, posted-by user, and confirmation reference	P1

6.6 Time Tracking

ID	Requirement	Priority
PUM-FR-050	PUM0060 shall record time spent per user per TSB stage: authoring, review, approval, posting	P1
PUM-FR-051	Time entries shall roll up to KPI reports: average time-to-publish per TSB type, per model	P1
PUM-FR-052	Time tracking data shall feed into the INT-06 outbound EDW pipeline for SAS VA analytics	P2

6.7 Audit Log

ID	Requirement	Priority
PUM-FR-060	All TSB actions (create, edit, submit, approve, reject, post, archive) shall be logged with: user ID, role, timestamp, action, previous and new value	P1
PUM-FR-061	Audit log shall be read-only; accessible to Admin and PQM roles only	P1
PUM-FR-062	Audit log shall support export to PDF and XLSX for NHTSA compliance reporting	P1

6.8 Master Data Management (PUM Admin)

ID	Requirement	Priority
PUM-FR-070	Admin shall manage model, model year, plant, and op code master data — primarily sourced from INT-01 with manual override capability	P1
PUM-FR-071	Admin shall manage parts master data — sourced from INT-04 (SAP BW) with manual override	P1
PUM-FR-072	Admin shall configure: TSB number format, approval chain templates, attachment limits, notification recipients	P1

6.9 Access Control

ID	Requirement	Priority
PUM-FR-080	Access to PUM screens shall be restricted by RBAC role: TSB authors (TE/DE), reviewers (QE/CE), approvers (ASM/DM/PQM), admins (Admin)	P1
PUM-FR-081	Non-KUS employees shall be grantable TSB submission access by Admin with a defined access expiry date	P1
PUM-FR-082	All access grants and revocations shall be logged in the audit trail	P1

7. Functional Requirements — Issue Management (ISM)

7.1 Issue Identification & Listing

ID	Requirement	Priority
ISM-FR-001	ISM0010 shall list all active quality issues with filter by: severity score, model, system, status, assigned role, date range	P1
ISM-FR-002	Issues shall be surfaced from five primary source channels: Warranty (INT-03 warranty claim threshold), Weibull (statistical reliability analysis), Comeback (repeat repair pattern from INT-03), Techline (dealer technical support inquiry), and FPQR (Field Product Quality Report via QIM0040). EWS flag (EWS_FG) signals from SAS/QC and GQIS inbound (INT-02 Group A) remain as additional signal sources. Each channel has a defined Phase 1 manual entry mode and a Phase 2 automation path — see §7.2.	P1
ISM-FR-003	Issue list shall display: issue ID, title, model, severity score, status, owner, days open	P1
ISM-FR-004	Issue list shall support bulk-select actions: assign, change status, export	P2

7.2 Issue Source Channels

Five primary channels trigger quality issue creation in N-PQMS. Each channel has a defined Phase 1 manual entry mode and a Phase 2 automation path.

Channel	Trigger	Phase 1 Mode	Phase 2 Automation
Warranty	Warranty claim volume exceeds threshold for model/system	Manual entry by QE/TE based on INT-03 claims data	INT-03 auto-threshold monitoring → auto-draft issue
Weibull	Reliability analysis identifies elevated failure rate for a population	Manual structured entry by QE (β , η , failure rate fields)	Automated Weibull computation from INT-03/INT-04 data
Comeback	Repeat repair on same VIN/symptom within rolling time window	Manual entry by QE triggered by INT-03 repair order review	INT-03 repeat repair auto-detect → auto-draft issue
Techline	Dealer technical support inquiry identifies a recurring quality pattern	Manual entry by QE/TE from Techline inquiry reports	INT-08 (TechLink) automated signal → issue draft
FPQR	Field Product Quality Report from field engineer flags a recurring issue	Manual promotion via QIM0040 → ISM link by QE	INT-07 (PQPS) automated FPQR ingest at volume threshold

ID	Requirement	Priority
ISM-FR-CH-001	Warranty Source: ISM0020 shall include a Warranty source entry form capturing: warranty claim count, threshold crossed, model, model year, system, date range of claims, and a link to the INT-03 warranty data view. Phase 1: manual entry by QE/TE after reviewing INT-03 data. Phase 2: INT-03 auto-threshold alert triggers draft issue creation.	P1
ISM-FR-CH-002	Weibull Source: ISM0020 shall include structured Weibull analysis input fields: Shape Parameter β , Scale Parameter η (characteristic life), failure rate at current mileage/time, sample population size, confidence interval. Phase 1: manual entry by QE. Phase 2: automated Weibull computation from INT-03/INT-04 data feeds the structured fields.	P1
ISM-FR-CH-003	Comeback Source: ISM0020 shall include a Comeback source entry form capturing: VIN range with repeat repairs, symptom code, number of return visits, time window (days), dealer region. Phase 1: manual entry by QE triggered by INT-03 repair order review. Phase 2: INT-03 repeat repair auto-detect → auto-draft issue.	P1
ISM-FR-CH-004	Techline Source: ISM0020 shall include a Techline source entry form capturing: inquiry reference ID, inquiry date, symptom description, affected model/system, number of dealers reporting the same issue. Phase 1: manual entry by QE/TE from Techline inquiry reports. Phase 2: INT-08 (TechLink) automated signal integration.	P1
ISM-FR-CH-005	FPQR Source: When an FPQR submitted via QIM0040 is flagged by QE as a recurring field quality issue, the system shall allow the QE to promote it to an ISM issue with source channel = FPQR and the originating FPQR ID auto-linked. Phase 1: manual promotion via QIM0040 → ISM link. Phase 2: INT-07 (PQPS) automated ingest triggers issue draft when FPQR volume threshold is crossed.	P1
ISM-FR-CH-006	The Issue Source field shall be mandatory on ISM0020. Valid values: Warranty / Weibull / Comeback / Techline / FPQR / EWS / GQIS / Manual. ISM0010 shall provide Issue Source as a filter field. Once set and saved, the source field is immutable except by ASM/PQM role with a justification note logged in the audit trail.	P1

7.3 Issue Entry & Scoring

ID	Requirement	Priority
ISM-FR-010	ISM0020 shall capture: issue title, description, model, model year, system, subsystem, symptom, affected VIN range, reported date, and Issue Source (mandatory dropdown: Warranty / Weibull / Comeback / Techline / FPQR / EWS / GQIS / Manual). Selecting a source shall reveal the channel-specific structured fields defined in §7.2.	P1
ISM-FR-011	ISM0030 shall display a severity score (0-100) calculated by the prototype scoring algorithm using: field frequency, repair cost, warranty claims count (from INT-03 Siebel). For Weibull-source issues, the scoring algorithm shall additionally ingest the structured Weibull parameters (β , η , failure rate) as severity inputs. Phase 2: AI-assisted scoring extended to all source channels.	P1
ISM-FR-012	Severity score shall be visible to QE/TE/DE roles; editable (with justification) by ASM/PQM roles	P1
ISM-FR-013	VIN-level data for an issue shall be auto-populated from INT-01 (AS400/HISNA): model, plant, production date, options	P1
ISM-FR-014	Dealer data for an issue shall be auto-populated from INT-03 (Siebel/DMS): dealer code, region, repair order count	P1

7.4 EWS Flag Integration

ID	Requirement	Priority
ISM-FR-020	N-PQMS shall expose <code>POST /api/ews/signals</code> to receive EWS alert signal flags from SAS/QC (Group D equivalent endpoint — internal)	P1
ISM-FR-021	On receiving a valid signal, N-PQMS shall set <code>EWS_FG = true</code> on the matching issue/QIR record	P1
ISM-FR-022	<code>EWS_FG</code> flag shall be displayed as a visual indicator on ISM0010 (issue list), ISM0020 (issue detail), ISM0130 (EWS Flag View), and all QIM screens	P1
ISM-FR-023	<code>POST /api/ews/signals</code> shall validate: authentication token (JWT), signal payload schema, VIN/model match; invalid payloads shall return HTTP 400 with error code	P1
ISM-FR-024	EWS signal receipt shall be logged in the audit trail with: signal ID, timestamp, model, VIN range, <code>EWS_FG</code> value set	P1

7.5 Issue Disposition

ID	Requirement	Priority
ISM-FR-030	ISM0070 shall record final disposition per issue: TSB (Technical Service Bulletin), SA (Service Action), SC (Safety Campaign), No Action, Monitoring	P1
ISM-FR-031	Selecting TSB disposition shall auto-create a PUM draft linked to the issue ID	P1
ISM-FR-032	Disposition shall require ASM or PQM role approval before it is recorded as final	P1
ISM-FR-033	GQIS outbound sync (INT-02 Group B) shall transmit disposition status updates to GQIS Korea HQ	P1

7.6 Issue Tracking & Lifecycle

ID	Requirement	Priority
ISM-FR-040	ISM0080 shall display the full lifecycle timeline of an issue: open → scored → QIR created → disposed → closed	P1
ISM-FR-041	State transitions shall be enforced by Camunda BPM — only valid transitions allowed per role	P1
ISM-FR-042	Days-open counter shall be visible on issue list and detail screens	P1
ISM-FR-043	Issues closed without disposition (>90 days old) shall trigger a reminder notification to the assigned QE	P2

7.7 Parts Request

ID	Requirement	Priority
ISM-FR-050	ISM0090 shall allow TE/DE to submit a parts request linked to an issue, specifying: part number (from INT-04 SAP BW), quantity, urgency	P1
ISM-FR-051	Parts request status shall be trackable: submitted → approved → ordered → received	P1

7.8 Communication Log

ID	Requirement	Priority
ISM-FR-060	ISM0100 shall maintain a threaded communication log per issue: internal comments, email correspondence, and cross-org messages	P1
ISM-FR-061	Communication entries shall be role-stamped and timestamped; deletion is not permitted (soft-hide only by Admin)	P1
ISM-FR-062	Outbound emails sent from N-PQMS related to an issue shall be automatically captured in the communication log	P2

7.9 Escalation

ID	Requirement	Priority
ISM-FR-070	ISM0110 shall allow escalation of an issue to: QIR status (creates QIM record), Top Issue status (flags for executive reporting)	P1
ISM-FR-071	Escalation to QIR shall trigger a Camunda BPM task assigned to the ASM role	P1
ISM-FR-072	Escalation to Top Issue shall trigger notification to PQM and Director roles	P1

7.10 Cross-Org Visibility

ID	Requirement	Priority
ISM-FR-080	ISM0120 shall allow issues flagged for cross-org sharing to be visible to non-KUS organizational units with read-only access	P1
ISM-FR-081	Cross-org sharing shall be grantable per issue by ASM/PQM roles; default is KUS-only visibility	P1

8. Functional Requirements — QIR Module (QIM)

8.1 QIR Lifecycle

ID	Requirement	Priority
QIM-FR-001	QIM0010 shall list all QIR records with filter by: status, model, system, severity, assignee, date created, FPQR-linked flag	P1
QIM-FR-002	QIM0020 shall display the full QIR record: issue origin, severity score, linked ISM issue, linked FPQRs, warranty claims (INT-03), VIN data (INT-01), communication log, disposition history	P1
QIM-FR-003	QIR creation shall be triggered by: escalation from ISM, direct creation by QE/TE role, inbound GQIS signal (INT-02 Group A)	P1
QIM-FR-004	QIR status lifecycle: Draft → Under Review → Approved → Closed / Escalated	P1
QIM-FR-005	QIR tracking dashboard (QIM0100) shall display: open QIRs by model, average resolution time, severity distribution, model-line breakdown	P1

8.2 QIR Approval Workflow

ID	Requirement	Priority
QIM-FR-010	QIM0030 shall present Camunda BPM task forms for multi-level QIR approval: QE → ASM → PQM	P1
QIM-FR-011	Approval actions: approve, reject with comments, request additional information	P1
QIM-FR-012	QIR approval decisions shall sync to GQIS Korea HQ via INT-02 Group B within 30 minutes of decision	P1
QIM-FR-013	QIR rejection shall return status to Draft with rejection reason visible in version history	P1

8.3 FPQR Submission & Management

ID	Requirement	Priority
QIM-FR-020	QIM0040 shall allow field engineers to submit an FPQR linked to an existing QIR or as standalone	P1
QIM-FR-021	FPQR fields: report date, dealer, VIN, mileage, symptom code, cause code, correction code, parts used, labor time, customer description	P1
QIM-FR-022	QIM0050 shall manage FPQR lifecycle: submitted → reviewed → closed; reviewer is QE/TE role	P1
QIM-FR-023	FPQR data shall feed into the warranty data analysis view (QIM0070) for trend analysis	P1

8.4 VIN Search

ID	Requirement	Priority
QIM-FR-030	QIM0060 shall enable search by VIN across INT-01 (AS400/HISNA) and INT-03 (Siebel/DMS)	P1
QIM-FR-031	VIN search results shall display: model, model year, plant, production date, warranty start date, open QIRs, repair history	P1
QIM-FR-032	VIN search shall return results within 3 seconds for single VIN lookup	P1

8.5 Analytics & Reporting (QIM)

ID	Requirement	Priority
QIM-FR-040	QIM0070 shall display warranty claims data (from INT-03 Siebel) aggregated by: model, system, date range — line charts, bar charts	P1
QIM-FR-041	QIM0080 shall compute QIR effectiveness: % of closed QIRs with verified corrective action; average days to close	P1
QIM-FR-042	QIM0090 shall display campaign reporting analytics: campaign issues by model, campaign closure rate	P1
QIM-FR-043	QIM0100 issue analytics dashboard shall include: issue volume trend, severity heatmap by model/system, open vs. closed ratio, EWS flag frequency	P1
QIM-FR-044	All QIM analytics data shall be included in the INT-06 outbound push to CDO	P1

9. Functional Requirements — Admin Module (ADM)

9.1 User & Access Management

ID	Requirement	Priority
ADM-FR-001	ADM0010 shall list all users with filter by: role, status (active/inactive/expired), org unit, last login date	P1
ADM-FR-002	ADM0020 shall allow Admin to create, edit, activate, deactivate, and delete user accounts; all changes logged in audit trail	P1
ADM-FR-003	ADM0030 shall allow Admin to assign and revoke RBAC roles per user; bulk assignment via XLSX import with validation	P1
ADM-FR-004	External user accounts (ADM0050) shall enforce an access expiry date; the system shall auto-deactivate accounts 24 hours after expiry with a notification to Admin	P1
ADM-FR-005	ADM0040 access audit log shall be read-only, immutable, exportable to PDF/XLSX; accessible to Admin and PQM roles only	P1
ADM-FR-006	Admin shall be able to force-expire a user's JWT session (effective within 1 token refresh cycle, maximum 30 minutes)	P1
ADM-FR-007	ADM0010 shall surface users with roles expiring within 14 days as a highlighted alert to Admin	P2

9.2 System Configuration

ID	Requirement	Priority
ADM-FR-010	ADM0060 shall allow Admin to configure global system parameters without code changes or redeployment	P1
ADM-FR-011	All configuration changes via ADM0060-ADM0100 shall be logged in the audit trail with: parameter name, old value, new value, user, timestamp	P1
ADM-FR-012	ADM0070 notification templates shall support: subject, body (with template variables: <code>{{user_name}}</code> , <code>{{issue_id}}</code> , <code>{{model}}</code>), recipient rules, trigger event	P1
ADM-FR-013	ADM0080 integration status monitor shall refresh every 60 seconds; errors shall be highlighted with error count, last-error message, and timestamp	P1
ADM-FR-014	ADM0090 batch schedule configuration shall allow Admin to change cron expressions for INT-06 EDW push without restart	P2

9.3 Workflow Template Management

ID	Requirement	Priority
ADM-FR-020	ADM0110 shall allow Admin to create, edit, activate, and deactivate Camunda BPMN workflow templates for TSB approval via a structured form or visual template editor	P1
ADM-FR-021	ADM0120 shall allow Admin to configure QIR assignment rules: model-to-QE affinity matrix, default assignee fallback	P1
ADM-FR-022	ADM0130 shall allow Admin to set SLA deadlines (hours) per workflow stage per template type; changes take effect on newly created workflow instances only	P1
ADM-FR-023	ADM0140 escalation rules shall define: condition (SLA exceeded, rejection count), escalation target (role or user), notification content	P1
ADM-FR-024	Workflow template activation/deactivation shall not affect in-flight workflow instances	P1

9.4 Master Data Management

ID	Requirement	Priority
ADM-FR-030	ADM0150-ADM0190 master data screens shall display: source system (INT-01 / INT-03 / INT-04), last sync timestamp, record count, sync status	P1
ADM-FR-031	Admin shall be able to trigger a manual on-demand sync for INT-01, INT-03, INT-04 master data from ADM screens	P1
ADM-FR-032	Manual overrides to master data shall be flagged with "Manual — Not from Source" label and included in audit trail	P1
ADM-FR-033	ADM0200 symptom/cause/correction code library shall be the single master used across ISM, QIM, and FPQR data entry	P1
ADM-FR-034	All master data changes shall trigger cache invalidation for the affected Redis keys (INT-01, INT-03, INT-04 caches)	P1

9.5 Batch & System Health

ID	Requirement	Priority
ADM-FR-040	ADM0300 batch job monitor shall display all scheduled jobs with: next-run time, last-run status (success/failure), duration, records processed	P1
ADM-FR-041	ADM0310 shall track Phase 1 dry-run migration progress (INT-11 sample); Phase 2 extends to full migration tracking	P1
ADM-FR-042	ADM0330 system health dashboard shall surface AWS CloudWatch metrics: EKS pod count/health, RDS connection pool utilization, Redis memory usage	P1
ADM-FR-043	Any batch job failure in ADM0300 shall trigger an Admin notification via SMTP relay (Group H at Phase 1; upgraded to INT-10 AWS SES in Phase 2)	P1

10. Authentication & Authorization

10.1 Single Sign-On (SSO)

ID	Requirement	Priority
AUTH-FR-001	N-PQMS shall integrate with HAEA SSO via Azure Active Directory using OAuth 2.0 / OpenID Connect	P1
AUTH-FR-002	All user sessions shall be managed via JWT tokens issued by Azure AD; token refresh interval configurable (default: 30 minutes)	P1
AUTH-FR-003	Session expiry shall redirect to HAEA SSO login page; unsaved work shall prompt the user before redirect	P1
AUTH-FR-004	KDealer Plus authentication flow (external dealer users) shall be supported via Azure AD B2C federation — configuration confirmed by HAEA IT (CP-4) before Architecture Design Week 1	P1

10.2 Multi-Factor Authentication (MFA)

ID	Requirement	Priority
AUTH-FR-010	MFA shall be enforced for all users via HAEA's Azure AD MFA policy; N-PQMS shall not bypass or override HAEA MFA settings	P1
AUTH-FR-011	MFA method selection (authenticator app, SMS, hardware key) is managed by HAEA IT; N-PQMS receives a valid MFA-passed JWT	P1

10.3 Role-Based Access Control (RBAC)

Role	Code	Description	Phase 1 Access
Quality Engineer	QE	Creates issues, scores, submits QIR	ISM + QIM full; PUM read
Test Engineer	TE	Creates issues, submits FPQR	ISM + QIM full; PUM read
Design Engineer	DE	Reviews issues, reviews QIR	ISM + QIM full; PUM read
Customer Experience	CE	Reviews issues, views QIR	ISM + QIM read; PUM read
After-Sales Manager	ASM	Approves QIR, approves disposition	ISM + QIM full; PUM approval
Deputy Manager	DM	Reviews and approves TSB	PUM approval; ISM + QIM read
Product Quality Manager	PQM	Final approval all modules	All Phase 1 screens full
Admin	Admin	System administration, user management, master data	All screens + admin

ID	Requirement	Priority
AUTH-FR-020	All N-PQMS screens and API endpoints shall enforce RBAC based on the JWT role claim from Azure AD	P1
AUTH-FR-021	Unauthorized access attempts shall return HTTP 403 with a user-visible error message; all attempts shall be logged	P1
AUTH-FR-022	Non-KUS external users shall be grantable access with a specific role and an expiry date; access shall auto-revoke on expiry	P1
AUTH-FR-023	Role assignments shall be managed via ADM0030 (Role Assignment screen) — fully functional at Phase 1 go-live; initial user population may be loaded via bulk XLSX import for go-live cutover	P1
AUTH-FR-024	All changes to user roles and access shall be logged in the audit trail	P1

10.4 Terms & Conditions

ID	Requirement	Priority
AUTH-FR-030	Users shall be required to accept Terms and Conditions on first login and whenever a revised version is published	P1
AUTH-FR-031	T&C acceptance shall be logged with: user ID, version accepted, timestamp	P1

11. Workflow Management (Camunda BPM)

11.1 Camunda BPM Engine — Phase 1 Scope

Phase 1 deploys the Camunda BPM engine for two core workflows. The full bidirectional INT-09 Camunda external API integration is Phase 2; Phase 1 uses embedded Camunda with internal task management.

ID	Requirement	Priority
WF-FR-001	N-PQMS shall deploy Camunda BPM (or equivalent, finalized during design phase) as the workflow engine for Phase 1 TSB and QIR workflows	P1
WF-FR-002	Camunda BPMN 2.0 process definitions shall be stored in GitLab and deployed via CI/CD pipeline	P1
WF-FR-003	Workflow state shall be persisted in the N-PQMS PostgreSQL database; workflow history shall be queryable by Admin and PQM roles	P1
WF-FR-004	All workflow transitions shall be logged in the N-PQMS audit trail	P1

11.2 TSB Approval Workflow (BPMN)

[Draft Created]
→ Technical Review (TE/DE)
 ↓ Reject → [Return to Draft]
 ↓ Approve
→ Manager Approval (ASM)
 ↓ Reject → [Return to Draft]
 ↓ Approve
→ Director Approval (DM/PQM)
 ↓ Reject → [Return to Draft]
 ↓ Approve
→ [Published]
 → Posting (PUM0050)
 → Archived

ID	Requirement	Priority
WF-FR-010	TSB approval workflow shall follow the BPMN chain above; each stage assignee determined by RBAC role mapping	P1
WF-FR-011	TSB workflow shall support configurable templates: standard, expedited (skip Director for minor revisions), and emergency (single-stage, PQM only)	P2
WF-FR-012	Each workflow stage shall enforce a configurable SLA deadline; overdue tasks shall escalate to the role supervisor	P2

11.3 QIR Assignment & Escalation Workflow (BPMN)

[Issue Escalated / QIR Created]
→ QIR Assignment (QE role auto-assigned by model/system rule)
 → Under Review (QE → TE → DE review)
→ QIR Approval (ASM)
 ↓ Reject → [Return to QE]
 ↓ Approve
→ PQM Final Approval
 ↓ Reject → [Return to ASM]
 ↓ Approve
→ [QIR Closed / Disposition Recorded]
 → GQIS outbound sync (INT-02 Group B)

ID	Requirement	Priority
WF-FR-020	QIR assignment shall be triggered automatically when an issue is escalated to QIR status in ISM	P1
WF-FR-021	QE assignee shall be determined by a configurable rule: model/system affinity matrix maintained by Admin	P1
WF-FR-022	QIR approval workflow shall mirror TSB approval: configurable multi-stage with rejection-return and escalation	P1

11.4 Workflow Templates & Configuration

ID	Requirement	Priority
WF-FR-030	Workflow templates shall be configurable by Admin role via a UI (no code change required)	P2
WF-FR-031	Role-to-stage mapping shall be configurable per workflow template	P2

12. Notifications & Alerts

ID	Requirement	Priority
NOTIF-FR-001	N-PQMS shall send email notifications via SMTP relay (Group H — <code>POST /api/mail/send</code>) for: task assignment, approval request, rejection, escalation, overdue SLA	P1
NOTIF-FR-002	Notification templates shall be configurable by Admin: subject, body, recipients, trigger event	P1
NOTIF-FR-003	All sent notifications shall be logged with: recipient, subject, trigger event, timestamp, delivery status	P1
NOTIF-FR-004	Notification failures (SMTP error) shall be retried up to 3 times with exponential backoff; persistent failures shall alert the Admin	P1
NOTIF-FR-005	In-app notification center shall display unread task assignments and approvals for the logged-in user	P2
NOTIF-FR-006	EWS alert receipt (<code>POST /api/ews/signals</code>) shall trigger a notification to the assigned QE for all issues with <code>EWS_FG</code> newly set to true	P1

Note: Full AWS SNS push notification (INT-10) is Phase 2. Phase 1 uses SMTP relay (Group H) for all email notifications.

13. Reporting & Dashboard (Prototype)

13.1 Standard Reports (12 Reports — Phase 1)

Reports are displayed directly in the N-PQMS UI (database-driven; not SAS VA). SAS VA dashboards consume the same data via INT-06 EDW outbound.

#	Report Name	Module	Key Metrics
R-01	TSB Publication Status Summary	PUM	TSBs by status, model, date range
R-02	TSB Approval Cycle Time	PUM	Avg. days per stage, bottleneck identification
R-03	TSB Time-to-Publish	PUM	By TSB type, model, author
R-04	Open Issue Summary	ISM	Issues by severity, model, system, age
R-05	Issue Disposition Summary	ISM	TSB / SA / SC / No Action counts by model
R-06	EWS Flag Report	ISM	Issues with EWS_FG = true by model, system, date
R-07	QIR Open/Closed Trend	QIM	Monthly QIR volume, open vs. closed
R-08	QIR Approval Cycle Time	QIM	Avg. days per stage
R-09	FPQR Volume by Dealer	QIM	FPQR count by dealer, model, date range
R-10	Warranty Claims by Model	QIM	Claim count, cost trend (from INT-03 Siebel)
R-11	Top Issues Dashboard	ISM + QIM	Top 10 open issues by severity score
R-12	User Activity Log	All	Actions per user, role, date range

13.2 Ad-hoc Reporting Structure (6 Types)

#	Ad-hoc Type	Description
A-01	Parameter-based QIR Query	User selects model, system, date range, severity range; system returns matching QIR records
A-02	VIN-level Drill-down	Enter VIN → all linked issues, QIRs, FPQRs, repair orders displayed
A-03	Custom TSB Filter Export	User selects filters → exports to XLSX
A-04	Cross-model Comparison	Side-by-side issue volume for up to 4 selected models
A-05	Date-Range Snapshot	All open items as of a specified historical date
A-06	Role-Activity Audit	Actions performed by a specific user or role over a date range

13.3 Prototype KPI Dashboard

ID	Requirement	Priority
RPT-FR-001	Prototype dashboard shall display: total open issues, total open QIRs, TSBs published YTD, avg. severity score (portfolio), EWS flag count	P1
RPT-FR-002	Dashboard KPIs shall be filterable by model and date range	P1
RPT-FR-003	Dashboard shall be accessible to ASM, DM, PQM, and Admin roles	P1
RPT-FR-004	Dashboard data shall refresh every 15 minutes (configurable by Admin)	P2

14. Integration Requirements — Phase 1

14.1 Integration Overview

INT	System	Direction	Protocol	Phase
INT-01	AS400 / HISNA	← Inbound	REST adapter (Group C — 13 endpoints)	P1
INT-02	GQIS Korea HQ	↔ Bi-directional	REST (Group A inbound 5 + Group B outbound 4)	P1
INT-03	Siebel / DMS	← Inbound	REST adapter	P1
INT-04	SAP BW / 4HANA	← Inbound	ODP/OData REST	P1
INT-05	SAP ERP	← Inbound	BAPI/RFC fallback REST	P1
INT-06	CDO (Redshift)	→ Outbound	Batch push (Parquet/S3 or Redshift load)	P1

14.2 INT-01 — AS400 / HISNA (Group C, 13 endpoints)

ID	Requirement	Priority
INT01-FR-001	N-PQMS shall consume INT-01 to retrieve: VIN master records, model codes, plant codes, dealer codes, parts data, op codes from the HISNA AS/400 KMAFPRD library	P1
INT01-FR-002	VIN lookups shall return results within 3 seconds for single VIN	P1
INT01-FR-003	INT-01 data shall be cached in Redis (AWS ElastiCache) with configurable TTL (default: 4 hours for master data) to reduce AS400 load	P1
INT01-FR-004	Credentials: HISNA AS/400 JDBC service account provided by CP-3 (ACC-07) by Jun 1, 2026	P1
INT01-FR-005	INT-01 adapter shall implement circuit breaker; on AS400 unavailability, return cached data with stale-data indicator	P2

14.3 INT-02 — GQIS Korea HQ (Groups A + B, 9 endpoints)

ID	Requirement	Priority
INT02-FR-001	Group A (Inbound — 5 endpoints): N-PQMS shall receive QIR data from GQIS via the IF_* staging tables (IF_GQISQIRH , IF_GQISQIRF , IF_GQISQISO , IF_GQISCMNT)	P1
INT02-FR-002	Inbound QIR sync shall run on a configurable schedule (default: every 30 minutes); new/updated GQIS records shall create or update matching N-PQMS QIR records	P1
INT02-FR-003	Group B (Outbound — 4 endpoints): N-PQMS shall transmit QIR status updates, disposition decisions, and approval records back to GQIS within 30 minutes of the triggering event	P1
INT02-FR-004	Bi-directional sync shall implement conflict resolution: GQIS is authoritative for QIR origin data; N-PQMS is authoritative for disposition and approval status	P1
INT02-FR-005	GQIS REST API must be available in the HAEA DEV environment by Jul 3, 2026 (ASSM-S5); if not, Group A/B acceptance scenarios are marked blocked	P1
INT02-FR-006	A 2-week parallel write period with GQIS HQ Korea (CP-1) is required before go-live to validate bidirectional sync accuracy	P1

14.4 INT-03 — Siebel / DMS (Inbound)

ID	Requirement	Priority
INT03-FR-001	N-PQMS shall consume INT-03 to retrieve: dealer master data (dealer code, name, region, contact), repair orders, warranty claims, K-Support cases	P1
INT03-FR-002	Warranty claims data from INT-03 shall feed: QIR severity scoring (ISM-FR-011), warranty analysis charts (QIM-FR-040), and INT-06 EDW outbound	P1
INT03-FR-003	INT-03 shall also serve WPC FixedOps data (Group F, 2 endpoints: GET /api/wpc/repair-orders , GET /api/wpc/parts/{partNo}) — per ASSM-07, WPC data is absorbed into INT-03 Siebel; standalone SQL Server adapter is not built unless ASSM-07 is found false	P1
INT03-FR-004	INT-03 connectivity is gated on N-GWMS AWS environment readiness (ASSM-01, Jun 10, 2026); fallback is AS400 warranty data (INT-01) for interim DEV testing	P1

14.5 INT-04 — SAP BW / 4HANA (Inbound)

ID	Requirement	Priority
INT04-FR-001	N-PQMS shall consume INT-04 to retrieve: parts master data (part number, description, price, supersession) and warranty cost data via ODP/OData interface	P1
INT04-FR-002	Parts data from INT-04 shall populate: PUM0020 parts field auto-complete, QIM0040 FPQR parts used field, parts request tracking (ISM-FR-050)	P1
INT04-FR-003	INT-04 data shall be cached in Redis with configurable TTL (default: 8 hours for parts master)	P2

14.6 INT-05 — SAP ERP (Fallback, Inbound)

ID	Requirement	Priority
INT05-FR-001	INT-05 shall serve as fallback for parts/material data when INT-04 SAP BW is unavailable, via BAPI/RFC REST interface	P2
INT05-FR-002	N-PQMS shall automatically switch to INT-05 fallback when INT-04 returns error or timeout > 10 seconds	P2

14.7 INT-06 — CDO Outbound

ID	Requirement	Priority
INT06-FR-001	N-PQMS shall implement an outbound batch job that pushes analytical data from AWS RDS PostgreSQL to CDO (Redshift or Parquet on S3) per the schema defined by the CDO team (ASSM-04, due Jul 14, 2026)	P1
INT06-FR-002	Tables included in Phase 1 INT-06 push: T_QIR , T_CLM_HDR (warranty claims), T_VIN_MASTER , T_DEALER , T_TSB , T_ISSUE , T_EWS_SIGNAL , T_FPQR , T_USER_ACTIVITY	P1
INT06-FR-003	INT-06 batch shall run on a schedule configurable by Admin (default: nightly at 2:00 AM local); incremental delta push based on updated_at timestamp	P1
INT06-FR-004	INT-06 push shall be confirmed by CDO team sign-off before Phase 2 Dev milestone (M7); Phase 1 Go-Live (M5) requires DEV-level INT-06 functional test only	P1
INT06-FR-005	INT-06 failure shall alert Admin via email; retry 3 times before alerting; data loss is not acceptable — failed batches shall be queued for re-run	P1

15. Security Requirements

ID	Requirement	Priority
SEC-FR-001	All N-PQMS interfaces shall use HTTPS (TLS 1.2 minimum); HTTP connections shall be redirected to HTTPS	P1
SEC-FR-002	All secrets (DB credentials, API keys, JWT signing keys) shall be stored in AWS Secrets Manager — plaintext <code>.properties</code> files are prohibited	P1
SEC-FR-003	SAST scan (SonarQube + OWASP dependency check) shall be run in the CI/CD pipeline on every merge to main branch; zero Critical findings permitted at M3 release	P1
SEC-FR-004	Code shall follow OWASP Top 10 secure coding guidelines; input validation enforced at all API boundaries	P1
SEC-FR-005	User passwords are managed by Azure AD — N-PQMS shall not store or process passwords	P1
SEC-FR-006	All changes to user data and access privileges shall be logged in an immutable audit trail (append-only)	P1
SEC-FR-007	PII data (VIN tied to owner, dealer employee info) shall be handled per HAEA/KMA PII policy; PII fields shall be masked in logs	P1
SEC-FR-008	N-PQMS shall comply with HAEA/KMA data retention policy; records older than the retention threshold shall be archivable by Admin (not auto-deleted)	P1
SEC-FR-009	Penetration testing shall be performed by HAEA after Phase 1 deployment; Provider shall deliver a PEN Test Readiness Report at Phase 1 SIT (M4)	P1
SEC-FR-010	AppScan / DAST testing shall be conducted by HAEA security team; Provider shall resolve Critical and High severity findings before go-live	P1
SEC-FR-011	Multi-tenant architecture shall be implemented (company code-based tenancy) to support future system extensions	P2
SEC-FR-012	Single-schema policy with tenant discriminator column shall be used for multi-tenant data isolation	P2

16. Non-Functional Requirements (NFR)

16.1 Performance — SPEC-PERF-001

ID	Screen / Endpoint	Metric	DEV Target	Staging Target
NFR-P-001	PUM0020 Publication Entry (NFR anchor)	Page load (hot cache)	≤ 2.0s @ 10 concurrent users	≤ 1.5s @ 50 concurrent users
NFR-P-002	All Tier 1 screens (PUM + ISM, 20 screens)	Page load	≤ 2.0s @ 10 users	≤ 2.0s @ 50 users
NFR-P-003	All Tier 2 screens (QIM, 10 screens)	Page load	≤ 3.0s @ 10 users	≤ 3.0s @ 50 users
NFR-P-004	VIN single lookup (INT-01)	API response	≤ 3.0s	≤ 3.0s
NFR-P-005	QIR list (first page, 20 records)	API response	≤ 2.0s	≤ 2.0s
NFR-P-006	System-wide concurrent users (Phase 1 go-live)	Sustained load	10 users (DEV)	50 users without >1% error rate

16.2 Availability & Reliability

ID	Requirement	Target
NFR-A-001	Production system availability (Phase 1 post-go-live, Hypercare period)	≥ 99.5% (excluding scheduled maintenance)
NFR-A-002	Scheduled maintenance window	Weekday nights 11 PM–2 AM PT; weekend all-day with 48h advance notice
NFR-A-003	RTO (Recovery Time Objective) — P1 incident	4 business hours
NFR-A-004	RPO (Recovery Point Objective)	1 hour (AWS RDS automated backup)

16.3 Scalability

ID	Requirement
NFR-S-001	EKS deployment shall support horizontal pod autoscaling (HPA) based on CPU/memory thresholds
NFR-S-002	Database connection pooling shall be configured to prevent connection exhaustion under load
NFR-S-003	Architecture shall support scaling to 200 concurrent users in Phase 2 without architectural changes

16.4 Code Quality

ID	Requirement	Gate
NFR-Q-001	Unit test coverage	≥ 75% line coverage at M3 (Phase 1 Dev)
NFR-Q-002	SonarQube quality gate	Zero Critical blockers; High issues < 10
NFR-Q-003	SAST scan	Zero Critical or High severity unresolved at M3
NFR-Q-004	API documentation	OpenAPI 3.0 YAML for all Phase 1 endpoints by M3

16.5 Browser & Device Support

ID	Requirement
NFR-B-001	Chrome (latest 2 major versions), Edge (latest 2), Safari (latest 2) on desktop
NFR-B-002	Responsive layout for tablet viewports (768px minimum width) — tested in BrowserStack (HAEA license)
NFR-B-003	Vue.js SPA shall load within 5 seconds on first visit (cold cache, standard enterprise network)

17. Data Migration — Phase 1 Scope

Phase 1 includes a **dry-run** of the INT-11 historical data migration. The full production migration (all TSB, QIR, FPQR records, ~300 GB+) is executed in Phase 2.

ID	Requirement	Priority
DM-FR-001	Provider shall deliver a Data Migration Dry-Run Report at M3 (Phase 1 Dev): sample migration of 5–10% of TSB and QIR records from legacy Oracle to PostgreSQL with reconciliation report	P1
DM-FR-002	Legacy KPQMS Oracle schema DDL (ACC-04) and production data extract for sizing shall be provided by HAEA by May 25, 2026	P1
DM-FR-003	Data migration approach shall use ETL scripts (Python/SQL) with: source → staging → target validation, row count reconciliation, and attachment binary transfer via S3	P1
DM-FR-004	User-visible historical TSB and QIR records (read-only view) shall be accessible in N-PQMS at Phase 2 go-live via INT-11 full migration	P2
DM-FR-005	Data quality rules shall be agreed with HAEA during M2 BRD workshops; records failing quality rules shall be flagged in the dry-run report, not silently dropped	P1

18. Assumptions & Dependencies

18.1 Phase 1 Critical Assumptions

ID	Assumption	Deadline	Risk if False
ASSM-01	N-GWMS AWS environment available to Provider	Jun 10, 2026	M3 (Nov 6) at risk; Change Request
ASSM-03	EWS alerting owned by KUS PQ/SAS team (Option A); PQMS scope = <code>POST /api/ews/signals</code> only	Jul 21, 2026	Change Request if PQMS-native EWS microservice directed
ASSM-07	WPC FixedOps data absorbed into INT-03 Siebel; no standalone SQL Server adapter	Jul 14, 2026	Change Request if standalone adapter required
ASSM-S3	Legacy KPQMS source code, JDBC credentials, WebLogic config, FSP maps (1,023 maps) provided	May 18, 2026	Discovery phase delayed day-for-day
ASSM-S5	GQIS Korea HQ REST API available in HAEA DEV environment	Jul 3, 2026	Group A/B blocked; acceptance scenarios deferred

18.2 Phase 1 HAEA Access Obligations

ACC #	Item	Deadline
ACC-01	Legacy KPQMS source code (WAR/EAR)	May 18, 2026
ACC-02	WebLogic configuration exports	May 18, 2026
ACC-03	FSP XML map exports (1,023 maps)	May 18, 2026
ACC-04	KPQMS Oracle schema DDL	May 25, 2026
ACC-05	GQIS <code>IF_*</code> staging table DDL	May 25, 2026
ACC-06	Batch job code (EwsViewBatch, WarrantySearchBatch, CampaignAlertBatch)	May 25, 2026
ACC-07	HISNA AS/400 JDBC credentials (CP-3)	Jun 1, 2026
ACC-08	Active Directory bind credentials (CP-4)	Jun 1, 2026
ACC-09	WPC SQL Server connection string (CP-7)	Jun 1, 2026
ACC-10	SMTP relay configuration (CP-8)	Jun 1, 2026
ACC-11	AWS N-GWMS environment access (VPC, IAM, EKS)	Jun 10, 2026
ACC-12	PQMS usage reports (screen hits, sessions)	May 25, 2026
ACC-13	GQIS REST API documentation (CP-1)	Jun 1, 2026
ACC-14	Tibero DB access (read-only, GQIS staging)	Jun 1, 2026
ACC-15	Legacy PQMS production access (read-only)	May 18, 2026

18.3 Environment Provisioning (HAEA Obligations)

Environment	Required By	Contents
DEV	Jul 14, 2026	EKS single-node, RDS PostgreSQL (dev), Redis, S3, VPC → HISNA AS/400
CI/CD Pipeline	Jun 30, 2026	GitLab, Jenkins, Argo CD, ECR, S3 artifact bucket
Staging	Oct 14, 2026	EKS prod-like, RDS prod-equivalent, full INT routing
Production	Dec 1, 2026	Multi-AZ EKS, RDS with backup, WAF, CloudWatch

19. Out of Scope — Phase 1

19.1 Deferred to Phase 2

- INT-07 PQPS/FPQR inbound · INT-08 TechLink · INT-09 Camunda bi-directional external API · INT-10 AWS SES/SNS · INT-11 full historical migration · INT-12 IQS/VDS
- ADM0210–ADM0250 (ML Model Management — 5 screens) · ADM0260–ADM0290 (IQS/VDS Data Management — 4 screens)
- Full AI/ML module (24 ML models)
- NLP / RAG intelligent search (full implementation)
- Enhanced analytics: anomaly detection, trend mining, duplicate issue detection
- AI severity auto-classification

19.2 Permanently Out of Scope

#	Item	Reason
OS-01	GBI Oracle DW adapter (<code>jdbc/GBIBiz</code>)	GBI decommissioned
OS-02	Spotfire/EWS Oracle adapter (<code>jdbc/SP0TBiz</code>)	Spotfire decommissioned
OS-03	<code>EwsViewBatch</code> Java batch or PQMS-native EWS microservice	SAS/QC (KUS PQ team) handles EWS
OS-04	Power BI embedded development	SAS VA/Viya selected
OS-05	SAS procurement / provisioning / licensing	HAEA/KUS PQ responsibility
OS-06	SAS VA dashboard / report authoring	KUS PQ team responsibility
OS-07	SAS/QC EWS job design and implementation	KUS PQ team responsibility
OS-09	CDO schema design or infrastructure	CDO responsibility
OS-12	Penetration testing	Performed by HAEA
OS-13	DevSecOps / CI/CD pipeline provisioning	Provisioned by HAEA
—	SAP ABAP / module changes	Always out of scope
—	Hardware, networking, physical infrastructure	Always out of scope

20. Phase 1 Milestone Schedule & Acceptance Criteria

20.2 M2 — Discovery & Planning (Jul 3, 2026) — BRD Gate

This BRD document is the primary artifact for M2-A1. Full acceptance criteria per Exhibit E §M2.

Must-pass for M2:

- ☐ All 12 INT workstream interfaces defined
- ☐ 27 active API endpoints confirmed (Groups A, B, C, F, G, H)
- ☐ Tier 1 screens (PUM + ISM = 20 screens) mapped to To-Be modules in BRD
- ☐ 34 Tier 4 screens documented as eliminated with user redirect path
- ☐ ASSM-01, ASSM-03, ASSM-05, ASSM-07 validation results documented
- ☐ UI/UX prototype covers PUM0020 and ≥ 5 additional Tier 1 screens
- ☐ Phase 1 delivery schedule confirmed (M3 Nov 6 baseline or revised with justification)

20.3 M3 — Phase 1 Dev & QA (Nov 6, 2026)

Scope: Tier 1 + Tier 2 + Tier 3 screens (PUM, ISM, QIM, ADM = 54 screens), INT-01-06, Camunda BPM, Azure AD SSO, EWS writeback, prototype dashboard, reporting structure.

Must-pass criteria:

- ☐ All 54 screens (PUM $\times$ 6, ISM $\times$ 14, QIM $\times$ 10, ADM $\times$ 24) functional and accessible on DEV
- ☐ ADM screens: ADM0010-ADM0100 (User Mgmt, System Config) fully functional; ADM0110-ADM0200 (Workflow Templates, Master Data) fully functional; ADM0300-ADM0330 (Batch/Health) fully functional
- ☐ Admin user able to create/edit user, assign roles, and trigger master data sync end-to-end via ADM screens
- ☐ PUM0020 Publication Entry — full create/edit/submit/approve/publish lifecycle functional
- ☐ TSB approval workflow (Camunda BPM) functional end-to-end on DEV
- ☐ QIR create/assign/resolve workflow (Camunda BPM) functional end-to-end on DEV
- ☐ Azure AD SSO login and session management functional
- ☐ RBAC enforced for all 8 roles (QE, TE, DE, CE, ASM, DM, PQM, Admin)
- ☐ INT-01 (AS400/HISNA) — VIN lookup returns valid data
- ☐ INT-02 (GQIS) — bidirectional QIR sync (Groups A + B) functional
- ☐ INT-03 (Siebel/DMS) — dealer lookup and warranty claims ingest functional (or stubbed with ASSM-S5 blocker documented)
- ☐ `POST /api/ews/signals` endpoint functional; `EWS_FG` flag surfaces on ISM/QIM screens
- ☐ ISM0020 Issue Source field mandatory and functional for all 5 primary channels: Warranty, Weibull, Comeback, Techline, FPQR — each channel-specific structured fields captured correctly
- ☐ ISM0010 Issue Source filter functional across all valid source values
- ☐ Weibull structured fields (β , η , failure rate, sample size) captured in ISM0020 and fed as inputs to ISM0030 severity scoring prototype
- ☐ FPQR $\rightarrow$ ISM promotion workflow functional (QIM0040 FPQR linked to ISM issue with source = FPQR)
- ☐ 12 standard reports structured and accessible; 6 ad-hoc report types functional
- ☐ **PUM0020 $\leq$ 2.0s response @ 10 concurrent users (DEV)**
- ☐ $\geq 75\%$ unit test line coverage across Phase 1 modules
- ☐ Zero Critical SAST findings; SonarQube zero Critical blockers

- ☐ Zero open P1/P2 severity defects at submission
- ☐ Data migration dry-run report delivered (5–10% sample, reconciliation report)

M3 Required Artifacts (from Exhibit E):

#	Artifact
M3-A1	Deployed application — DEV environment (HTTPS URL)
M3-A2	Source code — Phase 1 modules (GitLab)
M3-A3	Low-Level Design (LLD) — Phase 1
M3-A4	Database Schema — Phase 1 (SQL DDL + ERD)
M3-A5	Unit Test Results (JUnit HTML report)
M3-A6	Integration Test Results — INT-01 through INT-06
M3-A7	Code Coverage Report (JaCoCo)
M3-A8	API Documentation — Phase 1 endpoints (OpenAPI 3.0 YAML)
M3-A9	SonarQube Report
M3-A10	SAST Scan Report
M3-A11	Phase 1 UAT Plan
M3-A12	Known Issues Register
M3-A13	Data Migration Dry-Run Report

20.4 M4 — Phase 1 SIT (Nov 20, 2026)

Must-pass criteria:

- ☐ SIT pass rate $\geq$ 95% of Phase 1 test cases
- ☐ Zero open Critical severity defects
- ☐ All INT-01 through INT-06 exercised in end-to-end integration tests
- ☐ All 27 active API endpoints covered by SIT test cases (Groups A, B, C, F, G, H)
- ☐ BRD requirement traceability: $\geq$ 95% of Phase 1 functional requirements covered
- ☐ PEN Test Readiness Report delivered to HAEA security team

20.5 M5 — Phase 1 UAT & Go-Live (Dec 18, 2026)

Must-pass criteria:

- ☐ UAT sign-off received from Kim Yoonsoo (HAEA PM) in writing
- ☐ UAT pass rate $\geq$ 98% of Phase 1 UAT test cases
- ☐ Zero open Critical/High severity UAT defects at sign-off
- ☐ HAEA end-user training completed for QE/TE/DE/CE roles
- ☐ Production environment provisioned by HAEA (deadline Dec 1, 2026)
- ☐ Production deployment executed per Runbook — no unplanned rollback
- ☐ Post-deployment smoke test: all Tier 1 screens accessible, SSO login functional, PUM0020 workflow functional
- ☐ Legacy KPQMS cutover executed with HAEA sign-off

- ☐ Go-Live Readiness Checklist signed by Provider PM (Rita Verma) and HAEA PM (Kim Yoonsoo)
- ☐ CloudWatch monitoring confirmed operational on Production

20.6 M6 — Phase 1 Hypercare (Jan 15, 2027)

Hypercare SLAs (60 days post-go-live):

Severity	Response	Resolution
P1 — System down / data loss	1 business hour	4 business hours
P2 — Tier 1 screen non-functional	4 business hours	1 business day
P3 — Tier 2 screen issue	1 business day	3 business days
P4 — Minor / enhancement	2 business days	Change Order

Must-pass criteria:

- ☐ All P1 and P2 defects raised during Phase 1 Hypercare resolved or formally risk-accepted in writing by HAEA
- ☐ Hypercare Defect Log delivered
- ☐ Phase 1 knowledge transfer sessions completed (dates TBD at M5)

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Reference: SOW V0.9 DRAFT (2026-05-18) · HAEA-KUS_PQMS_Re-Platform_Implementation V0.9_DRAFT__20260518.docx

This BRD is deliverable M2-A1 (Exhibit E §M2). Detailed requirements will be validated and finalized during BRD workshops (May-Jun 2026) as part of Milestone 2 (Discovery & Planning Sign-off, Jul 3, 2026).